A Fuzzy TOPSIS Framework for Prioritising Pedestrian Crossing Safety and Accessibility Improvements

Soft Computing Assignment P20 | Indian Institute of Information Technology Kottayam

Abstract—This paper specifies an executable fuzzy TOPSIS decision-support model for ordering feasible pedestrian-crossing upgrade packages. The decision is intervention priority, not crash prediction or proof of treatment effectiveness. The supplied materials contained no site inventory or field observations, so empirical rankings remain pending. Four baseline criteria cover pedestrian demand, vehicle exposure, accessibility deficiency, and crossing-control or visibility deficiency; an extension adds verified essential-destination access. Observed count variation and paired assessor disagreement are approximated by triangular fuzzy numbers. The model uses Chen's vertex distance, summed across criteria, and a documented adaptation with crisp equal weights and attainable weighted ideals. A command-line pipeline creates base, crisp, extension, and weight-sensitivity results when observations are available. Synthetic records demonstrate only the arithmetic. Five test cases passed, including a separately calculated two-criterion example. The proposed campus study still needs a verified crossing inventory, repeated observations, engineering review, and independent validation.

Keywords—fuzzy TOPSIS, pedestrian crossings, accessibility, intervention prioritisation, decision support

1. Problem and research question

The decision is which feasible improvement package should receive attention first from campus facilities and the responsible road authority. Each alternative is one documented action package at a verified crossing, such as an access-ramp and tactile-cue repair or a control and visibility improvement. The proposed boundary is crossing points within the IIIT Kottayam campus and immediately adjoining public-road approaches. This boundary and the existence of candidate crossings require field confirmation. One package per crossing prevents the same site conditions being counted as separate competing actions. A higher closeness score means higher relative priority among the included packages; it is not an accident probability or proof that the action will reduce crashes.

Research question: under repeated field measurements and provisional equal weights, does adding verified essential-destination access change the leading feasible crossing-upgrade packages or the stability of the top group? The comparison tests a defined extension to the local deficiency-and-demand model. It does not claim that the extension improves decisions unless independent review supports that conclusion.

2. Related work

Study	Setting / method	Evidence and useful result	Limit for this project
Pedestrian-crossing safety decision support (2023), Transportation Research Procedia 69, 655–662; DOI 10.1016/j.trpro.2023.02.220	Checklist-based assessment; 10 unsignalized crossings on two Bologna roads	On-site checklist produced intervention priorities used in a municipal safety plan.	Local safety-rating instrument; not transferable as a universal India checklist.
Pedestrian Satisfaction-Based Methodology for Prioritization of Critical Sidewalk and Crosswalk Attributes Influencing Walkability (2021), Journal of Urban Planning and Development 147(3); DOI 10.1061/(ASCE)UP.1943-5444.0000718	Pedestrian satisfaction evidence with TOPSIS prioritisation	Highlights stop-line behaviour, turning conflicts, zebra markings and holding space in the reported study.	Ranks attributes by satisfaction, rather than individual capital interventions.
Evaluating the Pedestrian Gap Acceptance in Semicontrolled Midblock Crosswalks with an Integrated AHP-FL Approach (2022), Journal of Transportation Engineering Part A 148(9); DOI 10.1061/JTEPBS.0000720	Gap-acceptance analysis with AHP and fuzzy logic	Supports fuzzy treatment of imprecise crossing-behaviour judgments.	Different outcome and integrated method; not fuzzy TOPSIS.
Chen (2000), Extensions of the TOPSIS for group decision-making under fuzzy environment, Fuzzy Sets and Systems 114(1), 1–9	Triangular linguistic ratings; vertex distance and fuzzy ideals	Defines the fuzzy TOPSIS variant used here.	This project adapts its distance/closeness core to ranges in measured criteria; that adaptation is stated explicitly.

Jefmański and Sagan (2021) show that the choice of fuzzy conversion scale can materially change closeness values. This supports recording the evidence behind each triangular range and testing assumptions. It does not validate a particular scale for crossing audits; the proposed 0–3 rubric remains provisional.

FHWA guidance independently supports using pedestrian and vehicle volumes, crossing width and other variables when selecting treatments at unsignalized crossings; its thresholds are not transplanted to India. Visibility guidance identifies obstruction, lighting and roadway geometry as relevant visibility issues. Local engineering standards must determine which treatment is feasible.

3. Scope, criteria and collection

Identify each candidate by GPS position, road approach, crossing ID, photo evidence and proposed action. Include only a package that addresses a documented deficiency and passes a preliminary feasibility review; retain rejected proposals with reasons. Count crossing pedestrians and motor vehicles in matched 15-minute windows, with at least three intervals in morning, midday and evening periods on two ordinary weekdays. Record date, time, weather, events, direction and observer. Two assessors independently audit access and control/visibility features, retaining component observations and photo IDs. Reconcile differences only with written evidence. The count minimum, median and maximum become the triangle; assessor ratings supply the audit range. This schedule is a proposed field protocol, not completed research.

Criterion	Operational definition / unit	Direction and evidence	Uncertainty; overlap control
P1 Pedestrian demand	People crossing per hour; interval counts	Higher priority with more users; repeated manual counts	low/median/high from observed intervals; overlaps with destination counts, so destination criterion is access quality, not beneficiary count.
P2 Vehicle exposure	Vehicles passing per hour during same intervals	Higher exposure raises priority; same count windows	low/median/high across periods; related to P4 but represents exposure, not deficiency.
P3 Accessibility deficiency	0–3 structured access audit: curb transition, ramp, tactile cue, clear path, approach continuity	Higher indicates more severe unmet access; paired audit and photos	range covers assessor disagreement; assess components and record evidence; no duplicate population criterion.
P4 Safety-control / visibility deficiency	0–3 checklist of control appropriateness and observed sight/lighting/marking barriers	Higher indicates greater observed deficiency; paired audit	range from two assessors; keep distinct from traffic volume.
E1 Essential-destination access (extension)	0–3 evidence-based access importance score from mapped essential destinations within predeclared walk-network catchment	Higher priority when a deficient crossing limits essential access; verified mapping and walk audit	paired classification range; do not use destination count if correlated with demand; publish catchment and list.

The proposed rubric distinguishes no observed barrier (0), localized impairment (1), material obstruction or difficult detour (2), and an observed impassable approach or severe visible concern (3). For essential-destination access, the rating rises with the documented dependence of verified essential routes on the crossing. Component evidence, practical alternate routes and assessor disagreements must be recorded. The full provisional rubric is supplied separately for review before fieldwork.

The 0–3 ratings are operational study definitions, not an Indian compliance scale. Applicable local design rules and the actual action package must be reviewed by engineers before an eligibility decision. OpenStreetMap is a possible source for network context and crossing identification, but no local extract was obtained; map completeness would require site verification. Missing values remain missing and trigger re-observation. Raw interval and paired-audit forms, a collated decision matrix template, data dictionary and source register accompany the report. Pedestrian volume and destination access may overlap, so the extension records the consequence of losing an essential route rather than repeating beneficiary counts.

4. Fuzzy TOPSIS model

For alternative i and criterion j, collect a triangle $\tilde{x}_{ij}=(l,c,u)$, with $l \le c \le u$. For counts, these are the minimum, median and maximum observed hourly rates; for paired audits, they are evidence-supported lower, central and upper ratings. These spreads describe observed variation or assessor disagreement. A triangular membership function is zero outside [l,u], rises as $(x-l)/(c-l)$ to one at c, then falls as $(u-x)/(u-c)$; degenerate shoulders use limiting conventions. Membership is graded compatibility with the imprecise rating, not probability. Choosing a triangular shape is a transparent approximation; it is not calibrated without additional observations.

All selected criteria increase priority. For each j set Lj=min_i lij and Uj=max_i uij, then normalize each triangle component q by zijq=(xijq−Lj)/(Uj−Lj). Remove a criterion that is constant or identical across alternatives and renormalize the remaining nonnegative weights to sum to one; use wj=1/m provisionally. Form vij=wjzij. The project's attainable positive ideal is (wj,wj,wj) and its negative ideal is (0,0,0). With Chen's vertex metric d(a,b)=sqrt{[(al−bl)²+(ac−bc)²+(au−bu)²]/3}, sum criterion distances: Di+=Σj d(vij,Aj+) and Di−=Σj d(vij,Aj−). Closeness is Ci=Di−/(Di++Di−), ranked in descending order. The weighted ideals and observed-range normalization are explicit project adaptations to crisp weights and measured data; they are not presented as unchanged elements of Chen's group linguistic-rating example.

The implementation rejects unordered or nonfinite triangles, negative measurements, invalid weights, duplicate alternative IDs and zero denominators. An alternative set with no discriminating criterion cannot be ranked. Scaling bounds depend on the included candidates, so closeness scores are relative to that set. A cost criterion, if introduced later, must be reverse-oriented before this benefit-only model is used. Engineering feasibility is screened before ranking, while cost and budget constraints need a separate allocation decision.

5. Extension and evaluation plan

The extension adds E1, a 0–3 assessment of access to independently verified essential destinations through the crossing. Its walking-network catchment, destination list and alternative route evidence must be documented before scoring. Compare the same candidate set under P1–P4 and P1–P4+E1, with equal weights renormalized within each model. Compare fuzzy TOPSIS with a transparent crisp baseline: the equal-weight mean after a separate criterion-wise min–max normalization of central observations, omitting constant criteria. Report top-one change, top-three overlap and Spearman rank correlation. For weight sensitivity, multiply each equal weight by a sampled factor in [0.8,1.2], renormalize, and repeat 1,000 times with a recorded seed. Report how often each alternative ranks first and the top-three overlap. These measures describe model sensitivity, not validity. Independent validation requires a separate site audit or blinded engineering/accessibility review; none has yet been conducted.

Interpret the output as a screening order for technical review. Verify treatment appropriateness, dependencies, costs and budget fit separately; a rank ordering does not solve a budget-constrained portfolio problem. Do not infer crash reduction without an appropriate causal evaluation.

6. Implementation, reproducibility and current result status

The Python pipeline accepts a completed collated field CSV, validates numeric ranges and record identifiers, excludes documented infeasible candidates, saves a clean matrix and processing log, and writes the fuzzy baseline, crisp comparison, destination extension, sensitivity tables and an SVG chart. It was executed on a separately labelled synthetic dataset only. Five test cases passed, covering a hand-calculated two-criterion result, constant-criterion invariance, benefit/cost orientation, invalid data and end-to-end output. No actual crossing ranking, real-world sensitivity conclusion, extension benefit or independent validation can be reported from the supplied evidence. The exact commands and file schemas are in the README.

The contribution is a transparent, priority-oriented adaptation of fuzzy TOPSIS for crossing-upgrade packages, a field protocol tying fuzzy spreads to observed variation, and a matched test of essential-destination access. Chen's 2000 method supplies the vertex-distance and closeness core. The project's crisp weights, outer-bound normalization and weighted ideals are documented choices rather than claims about the original paper. Separate mathematical checks and reproducible output files allow these choices to be challenged in a viva.

7. Limitations and conclusion

No site inventory, repeated counts, paired audit, destination mapping, cost estimates or feasibility approvals were supplied. Equal weights may not reflect the decision-makers' priorities. Audit ratings and triangular spreads require consistency checks, and E1 may duplicate P1 if poorly defined. Rankings depend on candidate inclusion and normalization bounds. An agreement between methods would not establish ground truth. Once observations are independently checked, the score can support an initial order for technical and funding review, while treatment effectiveness and budget-optimal project selection remain separate questions.

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Appendix A. Worked example (hypothetical only)

For an independent two-criterion check, let hypothetical alternative A have normalized triangles (0.2,0.4,0.6) and (0.5,0.7,0.9), with weights 0.4 and 0.6. Its weighted triangles are (0.08,0.16,0.24) and (0.30,0.42,0.54); positive ideals are (0.4,0.4,0.4) and (0.6,0.6,0.6), while both negative ideals are (0,0,0). The positive vertex distances are $\sqrt{[(0.32^2+0.24^2+0.16^2)/3]}=0.2487301081$ and $\sqrt{[(0.30^2+0.18^2+0.06^2)/3]}=0.2049390153$. The negative distances are $\sqrt{[(0.08^2+0.16^2+0.24^2)/3]}=0.1728197520$ and $\sqrt{[(0.30^2+0.42^2+0.54^2)/3]}=0.4312771731$. Thus $D^+=0.4536691234$, $D^-=0.6040969250$ and $C=0.5711063670$. These values are asserted in the Python tests; they are hypothetical arithmetic, not field results.

For (0.2,0.4,0.6), membership at 0.3 is 0.5, at 0.4 is 1, and at 0.5 is 0.5. These values express compatibility with the assumed imprecise rating, not observed frequencies or probabilities.

Appendix B. Workflow

Figure B1. Project workflow from site verification to independent review.

No empirical ranking table appears because the required field observations do not exist in the supplied materials. All demonstration rankings and figures are explicitly marked hypothetical. The complete independent worked calculation also appears in docs/worked_example.md.

1. Verify crossing inventory and boundary

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2. Define feasible intervention alternatives

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3. Collect repeated counts and paired audits

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4. Clean and orient data → Fuzzy TOPSIS → baseline and extension comparison → sensitivity → independent review